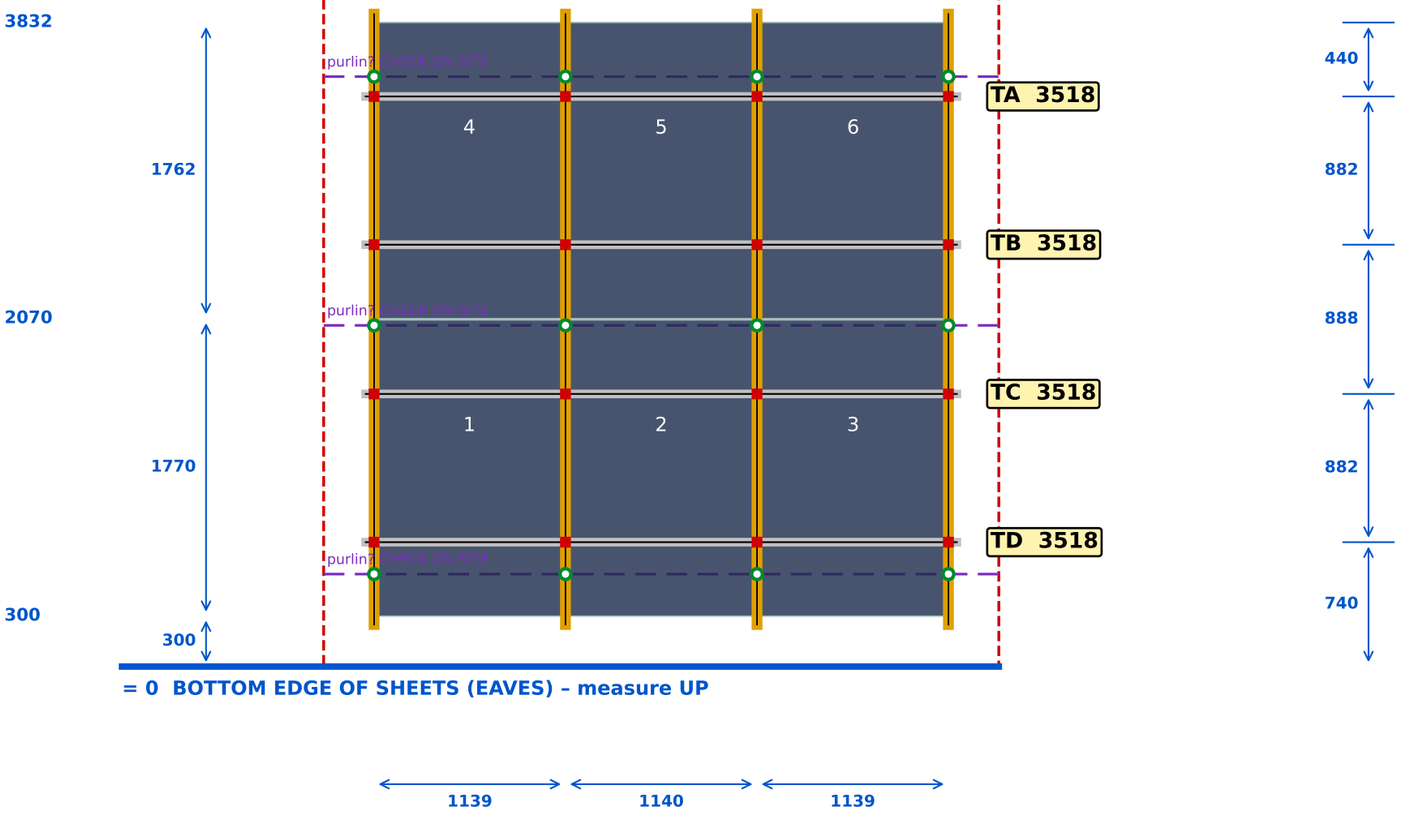
FINAL – 12 PANELS OUTBUILDING – PORTRAIT ON OVERLAY RAILS

LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | OUTBUILDING (12) – AC corrugated sheet on STEEL purlins | Install Wed 4 Nov 2026 | v5

2 faces × 6 | PORTRAIT 2 rows × 3 | array 3418 wide × 3532 up slope | BASE rails UP the roof on purlins, TOP rails ALONG roof on K2 cross connectors

Top of panels 3832 • RIDGE ≥ 4132 – CHECK ON SITE



= 0 **BOTTOM EDGE OF SHEETS (EAVES) – measure UP**

BASE rails B1…B4 across (centres from left panel edge) – blue mm

Orange = BASE rails UP the roof (3632, B1…B4), fixed at EVERY purlin (green ○ = 2003016 + 2002544). Grey = TOP rails ALONG roof (TA top…TD bottom), 2 per row, 440 in from panel ends. Red ■ = K2 cross connector (confirm part with Dale). Red tick = joiner. SAME BOTH FACES. Purple = purlins (nominal).

SET-OUT (up from = 0 at bottom edge of sheets)

Panels start 300. Row 1 300–2062, 8 gap, row 2 2070–3832. Base rails 250 → 3882.

Top rails TD 740 TC 1622 TB 2510 TA 3392 (gaps 740, 882, 888, 882, 440)

Slope needed ≥ 4132 vs ~3,800 estimate → 332 SHORT. Width ≥ 4018. CHECK ON SITE

⚠ AC SHEET ROOF – FRAGILE

- Crawl boards over ≥3 purlins + nets/harness. Never on sheets/rooflights.
- HSE a9: tape + damp (water/PVA spray), LOW speed, NO hammer, FFP3/P3.
- Tyvek 5/6, nitrile gloves, overshoes. Bag & label waste (red bags).
- **ELECTRICS = NOT OUR JOB (Regen sparks).**

BIRD MESH (if required – v2 said NO, confirm with Dale)

Each run = array width 3418 + 150 each end = 3718 mm

Top + bottom run per face × 2 faces = 4 runs × 3718 = 14.87 m

Cut list: 4 × 3718 mm (W-top, W-bottom, E-top, E-bottom)

FIXING – K2 2003016 Solarfastener 8×155/50 (STEEL)

1. Mark on CREST over purlin centre. Measure flange thickness.
2. **Sheet hole 11 mm, NO hammer. Steel 1.5–3 mm: self-drills. Thicker steel: pilot 6.8–7.4 mm. Clamp range 84–125 mm.**
3. Screw in at 90°, seal just compressed – don't over-drive.
4. 2002544 M10 adapter plate (1 per fastener) + T-bolt 1000571 + flange nut 1000042. Torque 32 Nm rail / 35 Nm joiners.
5. Fix every 1.0–1.2 m along each purlin.

PER FACE

BASE: 4 × 3632 up slope (1 bar each, ~1139 apart)

TOP: 4 × 3518 = 3518

Fixings 12 (4 × 3 purlins) Cross conn. 16

End clamps 8 Mid clamps 8 Joiners 0

LOAD LIST – 12 PANELS (both faces)

- ☐ **12 × JA JAM54D40-460/LB**
- ☐ **16 × K2 rail 3.65 m (8 base + 0 full top + 8 for 8×3518)**
- ☐ **0 × joiners ☐ 16 × end 30 mm ☐ 16 × mid 8 mm**
- ☐ **32 × K2 cross connectors – CONFIRM WITH DALE (not delivered?)**
- ☐ **24 × 2003016 fasteners (have 120 → 96 spare) ✓**
- ☐ **24 × 2002544 plates (have 85 → 61 spare) ✓**
- ☐ **24 × T-bolt 1000571 + 24 × flange nut 1000042**

TOOLS – tick when loaded

☐ Drill/driver + spare batts (NO hammer) ☐ 11 mm + 7 mm bits
☐ Sockets 13/15/17 + T30 ☐ Torque wrench 32/35 Nm
☐ Hacksaw + 24/32 TPI blades (alu), file/deburr
☐ Tape, chalk line, marker, masking tape, square
☐ WIRE CUTTERS ☐ STANLEY KNIFE ☐ Cable ties
☐ Sealant gun + sealant
☐ JSP Force 8 P3 / FFP3, Tyvek, gloves, overshoes
☐ Spray bottle water+PVA, red asbestos bags, labels, OB1 tape, wipes
☐ Crawl boards, harness/lanyards, edge protection
☐ **NO electrical tools – electrics are Regen's**

FLAGS – CHECK ON SITE

1. Count purlins each base rail crosses (3 assumed; 4 → 32 fixings).
2. Slope: needs ≥ 4132 vs est. 3,800 → ~330 SHORT: may not fit 300 mm rule.
3. Cross connectors not on delivery list – ring Dale before going up.
4. Flange > 3 mm → pilot 6.8–7.4 mm.

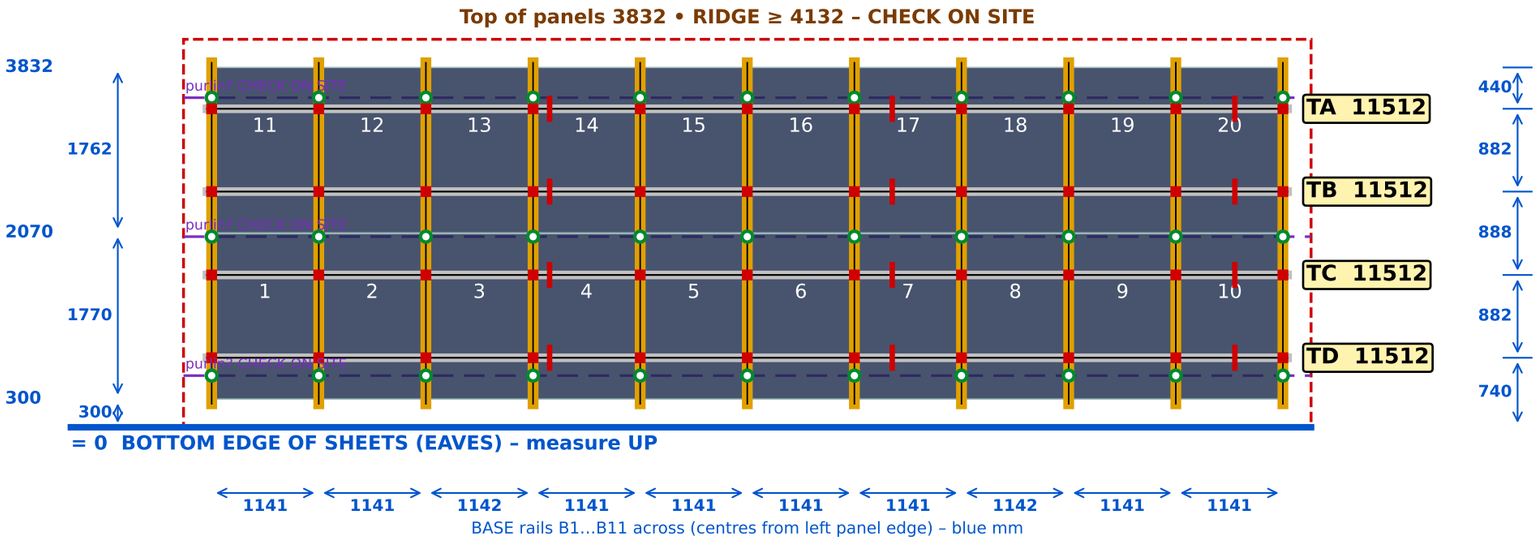
Top-rail cut bars: 3518 | 3518 | 3518 | 3518 | 3518 | 3518 | 3518 | 3518

Not for upload – John's info only. Not a structural design. Purlin-dependent dims marked CHECK ON SITE.

FINAL – 40 PANELS MAIN BARN – PORTRAIT ON OVERLAY RAILS

LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | MAIN BARN (40) – AC corrugated sheet on STEEL purlins | Install Wed 4 Nov 2026 | v5

2 faces × 20 | PORTRAIT 2 rows × 10 | array 11412 wide × 3532 up slope | BASE rails UP the roof on purlins, TOP rails ALONG roof on K2 cross connectors



Orange = BASE rails UP the roof (3632, B1…B11), fixed at EVERY purlin (green ○ = 2003016 + 2002544). Grey = TOP rails ALONG roof (TA top…TD bottom), 2 per row, 440 in from panel ends. Red ■ = K2 cross connector (confirm part with Dale). Red tick = joiner. SAME BOTH FACES. Purple = purlins (nominal).

SET-OUT (up from = 0 at bottom edge of sheets)

Panels start 300. Row 1 300–2062, 8 gap, row 2 2070–3832. Base rails 250 → 3882.

Top rails TD 740 TC 1622 TB 2510 TA 3392 (gaps 740, 882, 888, 882, 440)

Slope needed ≥ 4132 vs ~3,800 estimate → 332 SHORT. Width ≥ 12012. CHECK ON SITE

⚠ AC SHEET ROOF – FRAGILE

- Crawl boards over ≥3 purlins + nets/harness. Never on sheets/rooflights.
- HSE a9: tape + damp (water/PVA spray), LOW speed, NO hammer, FFP3/P3.
- Tyvek 5/6, nitrile gloves, overshoes. Bag & label waste (red bags).
- **ELECTRICS = NOT OUR JOB (Regen sparks).**

BIRD MESH (if required – v2 said NO, confirm with Dale)

Each run = array width 11412 + 150 each end = 11712 mm

Top + bottom run per face × 2 faces = 4 runs × 11712 = 46.85 m

Cut list: 4 × 11712 mm (W-top, W-bottom, E-top, E-bottom)

FIXING – K2 2003016 Solarfastener 8×155/50 (STEEL)

1. Mark on CREST over purlin centre. Measure flange thickness.
2. **Sheet hole 11 mm, NO hammer. Steel 1.5–3 mm: self-drills. Thicker steel: pilot 6.8–7.4 mm. Clamp range 84–125 mm.**
3. Screw in at 90°, seal just compressed – don't over-drive.
4. 2002544 M10 adapter plate (1 per fastener) + T-bolt 1000571 + flange nut 1000042. Torque 32 Nm rail / 35 Nm joiners.
5. Fix every 1.0–1.2 m along each purlin.

PER FACE

BASE: 11 × 3632 up slope (1 bar each, ~1141 apart)

TOP: 4 × 11512 = 3650 + 3650 + 3650 + 562

Fixings 33 (11 × 3 purlins) Cross conn. 44

End clamps 8 Mid clamps 36 Joiners 12

LOAD LIST – 40 PANELS (both faces)

- ☐ **40 × JA JAM54D40-460/LB**
- ☐ **48 × K2 rail 3.65 m (22 base + 24 full top + 2 for 8×562)**
- ☐ **24 × joiners ☐ 16 × end 30 mm ☐ 72 × mid 8 mm**
- ☐ **88 × K2 cross connectors – CONFIRM WITH DALE (not delivered?)**
- ☐ **66 × 2003016 fasteners (have 120 → 54 spare) ✓**
- ☐ **66 × 2002544 plates (have 85 → 19 spare) ✓**
- ☐ **66 × T-bolt 1000571 + 66 × flange nut 1000042**

TOOLS – tick when loaded

☐ Drill/driver + spare batts (NO hammer) ☐ 11 mm + 7 mm bits
☐ Sockets 13/15/17 + T30 ☐ Torque wrench 32/35 Nm
☐ Hacksaw + 24/32 TPI blades (alu), file/deburr
☐ Tape, chalk line, marker, masking tape, square
☐ WIRE CUTTERS ☐ STANLEY KNIFE ☐ Cable ties
☐ Sealant gun + sealant
☐ JSP Force 8 P3 / FFP3, Tyvek, gloves, overshoes
☐ Spray bottle water+PVA, red asbestos bags, labels, OB1 tape, wipes
☐ Crawl boards, harness/lanyards, edge protection
☐ **NO electrical tools – electrics are Regen's**

FLAGS – CHECK ON SITE

1. Count purlins each base rail crosses (3 assumed; 4 → 88 fixings).
2. Slope: needs ≥ 4132 vs est. 3,800 → ~330 SHORT: may not fit 300 mm rule.
3. Cross connectors not on delivery list – ring Dale before going up.
4. Flange > 3 mm → pilot 6.8–7.4 mm.

Top-rail cut bars: 562+562+562+562+562+562 | 562+562

Not for upload – John's info only. Not a structural design. Purlin-dependent dims marked CHECK ON SITE.

FINAL LOAD LIST – PORTRAIT OVERLAY – 12 + 40 PANELS

LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | MAIN BARN (40) – AC corrugated sheet on STEEL purlins | Install Wed 4 Nov 2026 | v5

Both systems: base rails UP roof on purlins (3 purlins assumed – CHECK ON SITE), top rails ALONG roof on K2 cross connectors

Item	12-panel	40-panel	TOTAL
JA JAM54D40-460/LB panels	12	40	**52**
K2 rail 3.65 m bars	16	48	**64**
Rail joiners	0	24	**24**
End clamps 30 mm	16	16	**32**
Mid clamps 8 mm	16	72	**88**
K2 cross connectors (CONFIRM with Dale)	32	88	**120**
2003016 Solarfastener 8×155/50	24	66	**90**
2002544 M10 adapter plates	24	66	**90**
M10 T-bolt 1000571	24	66	**90**
Flange nut 1000042	24	66	**90**
Bird mesh m (if required)	14.872	46.848	**61.72**

TOOLS – tick when loaded

☐ Drill/driver + spare batts (NO hammer) ☐ 11 mm + 7 mm bits

☐ Sockets 13/15/17 + T30 ☐ Torque wrench 32/35 Nm

☐ Hacksaw + 24/32 TPI blades (alu), file/deburr

☐ Tape, chalk line, marker, masking tape, square

☐ WIRE CUTTERS ☐ STANLEY KNIFE ☐ Cable ties

☐ Sealant gun + sealant

☐ JSP Force 8 P3 / FFP3, Tyvek, gloves, overshoes

☐ Spray bottle water+PVA, red asbestos bags, labels, OB1 tape, wipes

☐ Crawl boards, harness/lanyards, edge protection

☐ **NO electrical tools – electrics are Regen's**

vs DELIVERED

Fasteners: need 90, have 120 → +30

Plates: need 90, have 85 → -5

(delivery is for the 40 only? If 12-panel kit separate, 40-panel alone: 66 vs 120 / 85)

Cross connectors 120 – not on delivery list, ring Dale

Rails: 12-panel top 4 × 3518 (3518), base 8 × 3632; 40-panel top 8 × 11512 (3650+3650+3650+562), base 22 × 3632

40-panel slope needs ≥ 4132 vs ~3,800 est. – CHECK ON SITE

⚠ AC SHEET ROOF – FRAGILE

- Crawl boards over ≥3 purlins + nets/harness. Never on sheets/rooflights.
- HSE a9: tape + damp (water/PVA spray), LOW speed, NO hammer, FFP3/P3.
- Tyvek 5/6, nitrile gloves, overshoes. Bag & label waste (red bags).
- **ELECTRICS = NOT OUR JOB (Regen sparks).**